Dark Matter Project 2: XENON1T Approximate Nuclear-Recoil Likelihood (Recast)

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Abstract

We study a simplified dark-matter direct-detection problem using the public XENON1T approximate nuclear-recoil likelihood. The goal of the project is to constrain a standard spin-independent elastic WIMP–xenon recoil model and reproduce a limit-curve style result as a function of WIMP mass. To do this, benchmark recoil spectra were generated for WIMP masses between 5 and 200 GeV, passed through the XENON1T likelihood interface, and analyzed with a profile-likelihood scan over the signal-strength parameter. For each benchmark mass, an asymptotic 90% upper limit was extracted and collected into a limit-curve data file.

The resulting curve shows a strong mass dependence: the upper limit is very weak at low masses, decreases rapidly between 5 and 30 GeV, and then flattens to a few 10^{-3} in the normalization units used by the template spectra above about 50 GeV. The full workflow is reproducible from the repository and includes model generation, likelihood evaluation, and plotting. Because the recoil templates were exported with an additional numerical normalization factor for stability, the final curve should be interpreted as a limit on template normalization rather than a fully calibrated physical cross-section limit. Nevertheless, the project demonstrates a complete and working XENON1T recast pipeline.

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1 Introduction

Direct-detection dark matter experiments search for rare nuclear recoils produced by scattering of dark matter particles in underground detectors. In this project, the analysis is based on the public XENON1T approximate nuclear-recoil likelihood release and its associated data products [1–3].

The goal of the project is to use the public likelihood interface to constrain a chosen recoil-spectrum model and reproduce a limit-curve style result comparable, at least qualitatively, to the official XENON1T presentation. In the final version of the project, we use a simplified spin-independent elastic WIMP–xenon scattering model, generate recoil-spectrum templates for a set of benchmark WIMP masses, and compute asymptotic 90% upper limits on the corresponding signal-strength parameter.

This report is organized as follows. Section 2 summarizes the physical background. Section 3 describes the data products used in the final analysis. Section 4 presents the signal model. Section 5 explains the statistical method, including the likelihood treatment and the absence of explicit priors in the present workflow. Section 6 describes the inference procedure. Section 7 presents the final results and compares the reproduced curve qualitatively to the expected XENON1T behavior. The remaining sections discuss limitations, possible extensions, reproducibility, and author contributions.

2 Physical Background

XENON1T is a direct-detection dark matter experiment designed to search for weakly interacting massive particles (WIMPs) through their scattering off xenon nuclei [2]. In such a search, the signal is modeled as a recoil-energy spectrum, and the experiment constrains whether that spectrum is compatible with the observed data.

For the purposes of this project, the relevant observable is the differential recoil spectrum

$$\frac{\mathrm{d}R}{\mathrm{d}E}$$

where E is the recoil energy in keV. This quantity describes the expected event rate as a function of deposited recoil energy and is the object passed to the XENON1T approximate likelihood interface.

In a realistic dark-matter analysis, the spectrum depends on both the particle-physics interaction model and the astrophysical halo assumptions. Here we restrict attention to the standard spin-independent elastic case and treat the resulting recoil spectra as input templates for the XENON1T recast.

3 Data

The public data source used in this project is the XENON1T approximate nuclear-recoil likelihood release [3]. The final analysis no longer relies on the development toy spectrum used earlier for pipeline checks. Instead, the likelihood is evaluated on a set of physically motivated recoil-spectrum CSV files generated for benchmark WIMP masses

$$5,\ 10,\ 20,\ 30,\ 40,\ 50,\ 70,\ 100,\ 150,\ 200\ \text{GeV}.$$

Each spectrum file contains two columns:

- recoil energy,
- differential recoil rate.

These files are used as direct inputs to the XENON1T likelihood pipeline. The final processed output of the multi-mass run is collected in `data/processed/limit_curve.csv`, which stores the benchmark mass and the corresponding 90% upper limit obtained from the likelihood analysis.

During development, a toy spectrum in `data/raw/test_spectrum.csv` was also used to validate the workflow. That validation input is retained only for diagnostic purposes and is shown in the appendix.

4 Model

For the model part of the project, the standard spin-independent (SI) elastic WIMP–xenon scattering model was chosen, which was also the starting point suggested in the project brief. The task of the model implementation was to choose a signal model and compute the expected recoil spectrum to be passed to the XENON1T likelihood pipeline.

The recoil spectrum was implemented in `make_physical_model_csvs.py`. Spectra for benchmark WIMP masses of 5, 10, 20, 30, 40, 50, 70, 100, 150, and 200 GeV were generated and saved as CSV files with two columns: recoil energy and differential rate. The recoil-energy grid used in the implementation runs from 1 to 70 keV in steps of 1 keV.

The model includes the main parts of a simple SI recoil calculation: the WIMP–proton and WIMP–nucleus reduced masses, a Helm nuclear form factor, and a Standard Halo Model velocity integral $\eta(v_{\min})$. For the halo model, the benchmark values $\rho_\chi = 0.3$ GeV/cm^3, $v_0 = 220$ km/s, $v_E = 232$ km/s, and $v_{\rm esc} = 544$ km/s were used. A reference SI cross section of 10^{-45} cm^2 was used in the spectrum generator.

A practical implementation issue was that the raw physical rates were numerically very small. In the final implementation, a normalization factor of 10^{45} was introduced only for numerical convenience. This factor is not a physical unit conversion and was used only to keep the output values in a numerically convenient range. Therefore, the exported $\mathrm{d}R/\mathrm{d}E$ values should be interpreted as normalized recoil-spectrum templates rather than absolute physical recoil rates.

The main assumptions of this model are that scattering is elastic and spin-independent, xenon is represented by an effective nucleus with $A = 131$, and the dark-matter halo follows the Standard Halo Model. The main limitations are that the implementation is simplified and should be understood as a practical SI recoil-spectrum model rather than a fully precise detector-level prediction with complete unit treatment, isotope weighting, or alternative interaction types.

5 Statistical Method

The statistical framework used in this project is a profile-likelihood style scan over a signal-strength parameter multiplying the modeled recoil-spectrum templates. The modeled spectra are passed to the public XENON1T approximate likelihood interface, which evaluates how compatible each template is with the observed nuclear-recoil search data.

5.1 Likelihood

In the current code, the main wrapper function is

$$\texttt{BinwiseInference.from_xenon1t_sr(spectrum=spectrum).} \tag{1}$$

This constructs the XENON1T inference object from a chosen input spectrum. The script `src/io/run_likelihood_scan.py` provides the reusable single-spectrum interface and evaluates the likelihood-ratio function on the package default grid for a chosen input spectrum. For the final project result, the benchmark SI spectra were processed with the multi-mass runner `run_all_model_csvs.py`, which applies the same underlying XENON1T interface to each benchmark spectrum and prints the corresponding asymptotic 90% upper limit.

5.2 Priors

No explicit Bayesian prior specification is implemented in the inspected analysis path. The final workflow is therefore more accurately described as a profile-likelihood style scan with asymptotic upper-limit extraction, rather than a completed Bayesian posterior analysis.

6 Inference Procedure

The inference pipeline was run separately for each benchmark mass. The workflow is:

1. load one modeled recoil spectrum from a CSV file in `data/raw/`,
2. construct a `BinwiseInference` object,
3. evaluate the likelihood-ratio scan over the signal-strength parameter,
4. compute the asymptotic 90% upper limit using the χ^2_1 threshold,
5. repeat for all benchmark masses and collect the resulting upper limits in `data/processed/limit_curve.csv`.

The upper limit is computed by the function `safe_compute_ul`. When `asymptotic=True` is used, the code subtracts the minimum of the likelihood-ratio function and solves for the point at which the increase matches the one-degree-of-freedom asymptotic threshold. This produces one 90% upper limit for each benchmark mass. The reusable single-spectrum interface is implemented in `src/io/run_likelihood_scan.py`, while the benchmark mass sweep was run with `run_all_model_csvs.py`. The final file `data/processed/limit_curve.csv` is the committed saved output of that multi-mass workflow.

7 Results

The final analysis produces a mass-dependent upper-limit curve from 5 to 200 GeV. The numerical results are listed in Table 1. These values correspond to the upper limit on the template normalization used in the recoil-spectrum generation.

WIMP mass [GeV]	90% upper limit
5	427.9972
10	0.66713
20	0.015307
30	0.0056952
40	0.0040356
50	0.0034069
70	0.0028192
100	0.0024647
150	0.0023540
200	0.0024205

Table 1: Upper limits obtained from `data/processed/limit_curve.csv`.

7.1 Parameter Constraints

The strongest limits in this normalization scheme occur at higher masses, with the minimum value reached near 150 GeV. The curve weakens dramatically at very low mass, especially at 5 GeV, which reflects the reduced sensitivity of the detector to such recoil spectra in the simplified setup used here.

7.2 Model Comparison With Data

Figure 1 shows the final mass-dependent limit curve obtained from the benchmark spectra. The general trend is physically reasonable for a simplified direct-detection recast: sensitivity improves rapidly from very low mass to intermediate mass, then flattens in the higher-mass region. This is qualitatively consistent with the expected behavior of official direct-detection exclusion curves, which also lose sensitivity at low mass because the recoil energies fall closer to or below the detector threshold.

8 Discussion

The project now achieves the main computational goal of DM2: a working XENON1T recast pipeline that evaluates the public approximate likelihood on a chosen WIMP recoil model and returns a mass-dependent 90% limit curve. In that sense, the coding side of the project is complete.

The main remaining limitation is interpretational rather than technical. Because the modeled recoil spectra were rescaled by a large numerical normalization factor for stability, the final results are most safely interpreted as limits on template normalization, not yet as final physical cross-section limits in standard units. This does not invalidate the workflow, but it does limit how directly the current curve can be compared numerically to the official XENON1T exclusion curve.

A second limitation is that the model is intentionally simplified. It uses an effective xenon nucleus, a Standard Halo Model, and a restricted SI elastic interaction. Detector-level subtleties, isotope weighting, and more precise unit handling are outside the scope of the present implementation.

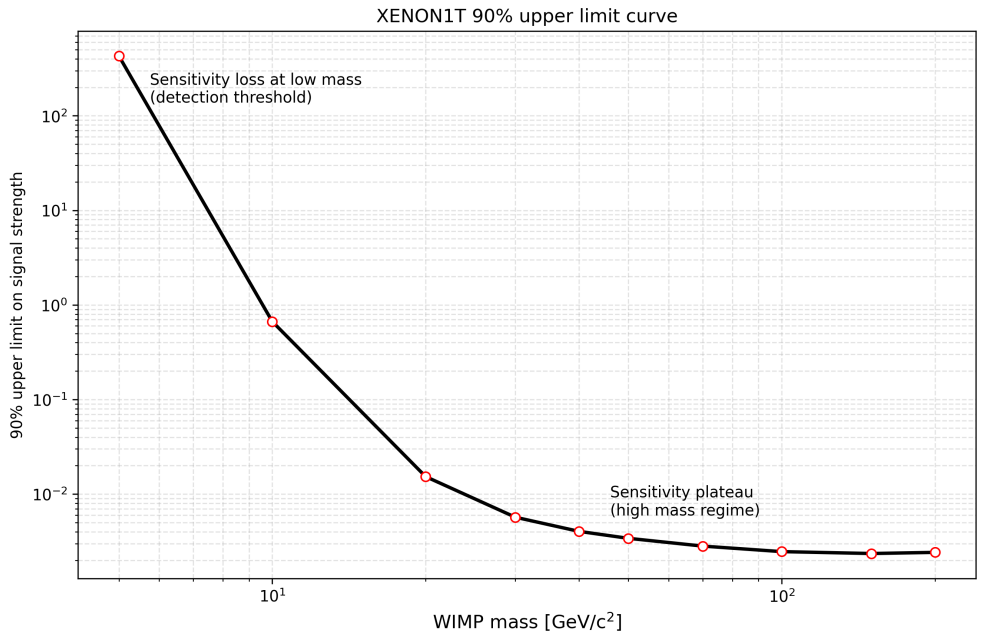

Figure 1: Mass-dependent 90% upper limit obtained from the XENON1T approximate likelihood for the benchmark SI elastic WIMP spectra used in this project. Because the recoil templates were exported with an arbitrary numerical normalization factor, the vertical axis should be interpreted as a template-normalization limit rather than a fully calibrated physical cross section.

9 Possible Extensions

A natural next step would be to remove the arbitrary normalization and convert the final upper limits into a fully calibrated cross-section curve. Another extension would be to compare the resulting curve directly to the official XENON1T result on the same axes. It would also be interesting to test alternative recoil models, such as different form factors or non-standard interactions, and to study how robust the curve is under changes in halo assumptions.

10 Conclusion

This project produced a complete first-pass recast of the XENON1T approximate nuclear-recoil likelihood for a simplified SI elastic WIMP model. Benchmark recoil spectra were generated for masses from 5 to 200 GeV, passed through the public XENON1T likelihood interface, and used to extract asymptotic 90% upper limits on the template normalization.

The final result is a mass-dependent limit curve with the expected qualitative behavior: very weak sensitivity at low masses and a much stronger, nearly flat constraint in the tens to hundreds of GeV range. Although the present normalization does not yet allow a clean one-to-one interpretation as a physical cross-section exclusion curve, the project successfully demonstrates the intended analysis pipeline and provides a solid basis for future refinements.

Author Contributions

- Floris Ordelmans (Data lead): set up the XENON1T approximate nuclear-recoil likelihood interface in the project repository, identified the relevant likelihood data products,

and built the code needed to read spectrum inputs and evaluate the likelihood. He also implemented the reusable pipeline that takes the model CSV spectra as input and produces the likelihood-scan and upper-limit outputs used by the rest of the analysis.

- Cristian Ioniță (Inference lead): contributed with code to the likelihood pipeline, ran it on the modeled spectra and computed the 90% upper limits on the signal strength for the benchmark masses. He also prepared the notebooks, helped organize the repository and made minor corrections to the report.
- Ivan Iturralde (Model lead): selected and implemented the physical signal model used in the project, namely the spin-independent elastic WIMP–xenon recoil-spectrum model, and generated the CSV spectra used as input for the likelihood analysis.
- Gianny Capitein (Plot/diagnostics lead): built the plotting scripts for the diagnostic figures and the final limit curve, improved the plot presentation, and checked that the resulting mass dependence behaved sensibly.
- Yuval Rochman (Integration/writing lead): coordinated the report, integrated the different parts into the final Overleaf document, aligned the write-up with the report structure, checked consistency between the repository outputs.

References

[1] E. Aprile et al. "An approximate likelihood for nuclear recoil searches with XENON1T data". In: *European Physical Journal C* 82 (2022), p. 989.

[2] E. Aprile et al. "Dark Matter Search Results from a One Tonne×Year Exposure of XENON1T". In: *Physical Review Letters* 121 (2018), p. 111302.

[3] XENON1T Collaboration. *XENON1T approximate nuclear recoil likelihood*. `https://github.com/XENON1T/xenon1t_approximate_nuclear_recoil_likelihood`. 2022.

Appendices

Additional Figures

The figures below are development-stage diagnostics retained for reproducibility and pipeline validation. They were useful during implementation, but they are not the main final physics results of the project.

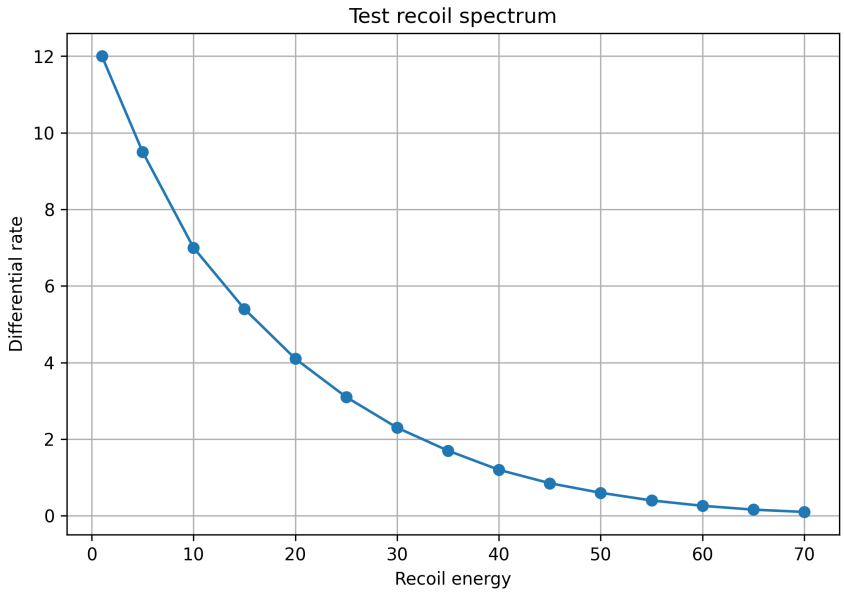

Figure 2: Toy recoil spectrum used early in development to validate the input pipeline.

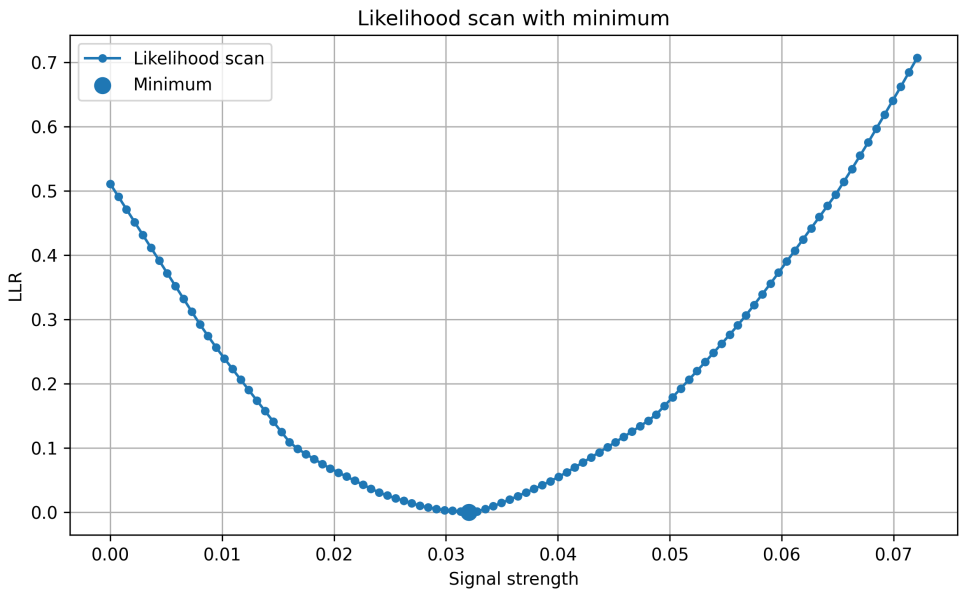

Figure 3: Example likelihood-ratio scan from the development stage, used to verify that the scan and minimum-finding workflow behaved as expected.

Reproducibility

The repository for this project is:

`https://github.com/FlorisOrd/stats-project-DM2`

The current repository includes a `requirements.txt` file with the packages `numpy`, `scipy`, `jupyter`, `matplotlib`, and `pandas`. A virtual environment can be created with

```
python -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip setuptools wheel
python -m pip install -r requirements.txt
```

The external XENON1T approximate likelihood package must also be installed in the environment:

```
cd ..
git clone https://github.com/XENON1T/xenon1t_approximate_nuclear_recoil_likelihood.git
cd xenon1t_approximate_nuclear_recoil_likelihood
python -m pip install .
cd ../stats-project-DM2
```

The main outputs used in the final report are:

- `data/processed/limit_curve.csv`
- `plots/limit_curve.png`

The main scripts currently used in the project are:

- `make_physical_model_csvs.py`, which generates the benchmark SI recoil-spectrum CSV files,
- `run_all_model_csvs.py`, which evaluates the benchmark spectra and prints the asymptotic 90% upper limits,
- `src/io/run_likelihood_scan.py`, which provides the reusable single-spectrum likelihood-scan example,
- `plot_check/plot_limit_curve.py`, which reads `data/processed/limit_curve.csv` and generates `plots/limit_curve.png`.

The final file `data/processed/limit_curve.csv` is included in the repository as the saved output used for the final report. The toy spectrum and annotated likelihood-scan outputs are retained only for development and diagnostic purposes.

Feedback

The project was interesting and manageable overall; the most challenging part, however, was connecting the physical recoil-spectrum model to the XENON1T likelihood pipeline in a numerically stable way and interpreting the final limits correctly. It would have been particularly helpful to have a somewhat clearer indication regarding what level of physical interpretation was expected in the final result—for example, whether a template-normalization limit was sufficient or whether a fully calibrated cross-section limit was expected. A small worked example showing the intended end-to-end workflow would have also helped reduce ambiguity. Overall, in terms of feasibility, we would assign the project a grade of 7.5/10—certainly doable, with some nontrivial implementation and interpretation issues.